# Ananya R

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## Summary

B.Tech Computer Science student at RV University with hands-on experience in NLP, speech AI and multimodal deep learning. Built a speech emotion recognition system (76.31% accuracy) and a multilingual abuse detection API (82% accuracy), and applied explainable AI to Alzheimer's detection (92.5% accuracy). Eager to build practical ML software.

## Technical Skills

**Languages:** Python, Java, C, SQL

**AI/ML:** Machine Learning, Deep Learning, Natural Language Processing (NLP), Speech Emotion Recognition, Affective Computing, Computer Vision, Feature Extraction, Fine-tuning, Hyperparameter Tuning, Explainable AI (XAI), SHAP

**Models:** Transformer, BERT, RoBERTa, XLM-RoBERTa, Wav2Vec2, WavLM, CNN, RNN, LSTM

**Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, Librosa, Flask, NumPy, Pandas, Matplotlib

**Tools & Concepts:** Git, GitHub, Google Colab, VS Code, Data Structures & Algorithms, DBMS, HTML, CSS, JavaScript

**Spoken Languages:** English (Professional), Kannada (Native)

## Experience

**Summer Intern – Agriculture AI**

*Dhee Centre for AI and Data Science, RV University* | Bengaluru, India | Jun 2025 – Jul 2025

- Built a Corn Leaf Disease Detection system using ML and deep learning to classify leaf images into Healthy, Rust, Blight, and Gray Leaf Spot, reaching **93%** (PRF+SVM) and **93.2%** (custom 4-layer CNN) accuracy.
- Extracted HSV, HOG, LBP, GLCM, Colour Histogram, and ORB features for Random Forest, SVM, KNN, Decision Tree, and PRF+SVM models; compared a custom CNN with VGG16, MobileNetV2, EfficientNet, InceptionV3, and ResNet152 using data augmentation, hyperparameter tuning, and accuracy/precision/recall/F1 evaluation.

## Projects

**Emotion Trend Analysis in Conversations Using Multimodal Audio and Text Data** (2026)

**Tech Stack:** Python, PyTorch, Wav2Vec2, WavLM, RoBERTa, BERT, Cross-Attention, IEMOCAP, Librosa, Matplotlib, SciPy

- Engineered a multimodal deep learning system on the IEMOCAP dataset for 4-class speech emotion recognition (angry, happy, sad, neutral), fusing audio and text features.
- Achieved **76.31% accuracy** with a Wav2Vec2 + RoBERTa-CNN + Cross-Attention fusion architecture.
- Benchmarked 3 unimodal and 8 multimodal model combinations (WavLM, DistilBERT, BERT-BiLSTM, CNN+BiLSTM); resolved class imbalance via class-weighted loss and excited-happy label merging.

**Multilingual Abuse Detection in English–Indian Code-Mixed Languages** (2026)

**Tech Stack:** Python, XLM-RoBERTa, HuggingFace Transformers, Flask, REST API, Sarvam AI STT

- Engineered a multimodal hate speech and abuse detection system for Tamil–English, Hindi–English, Kannada–English, and Malayalam–English code-mixed inputs via text and real-time speech, achieving **82% accuracy** on Tamil–English.
- Fine-tuned XLM-RoBERTa on the Dravidian Code-Mix 2020 and Hindi–English hate speech datasets using class-weighted binary cross-entropy loss.
- Integrated Sarvam AI Saarika v2.5 speech-to-text for real-time audio transcription through a Flask REST API backend.

**Early Alzheimer's Detection using Multimodal Deep Learning and XAI** (2025)

**Tech Stack:** Python, PyTorch, Bi-LSTM, MFCC, SpecAugment, TF-IDF, Scikit-learn, SHAP, LIME

- Built a multimodal Bi-LSTM fusing text transcripts and MFCC audio features to classify Control vs Dementia on 200 speech samples, achieving **92.5% accuracy** and **0.96 AUC**, outperforming the best classical ML baseline (SVM-RBF, 78.79%) and prior multimodal studies (85.3–88.7%).
- Applied SHAP, LIME, Bag-of-Words, and MFCC-spectrogram explainability to surface word- and acoustic-level cues (filler words, pauses, pitch variation), improving clinical interpretability.

## Education

**RV University** | Bangalore, India | *B.Tech in Computer Science and Engineering* | **CGPA: 8.82** (Expected 2027)
**Government Pre-University College, Amrutha** | Shivamogga, India | PCMB – **90.52%** (2023)
**Government High School Humchadakatte** | Shivamogga, India | SSLC – **86.72%** (2021)

## Certifications & Achievements

- **NPTEL Certifications (Elite + Silver):** Affective Computing; Introduction to Graph Theory
- **Finalist, Image AI-thon – Prompt Engineering Challenge:** Spring Fiesta, IIIT Surat (2026)